

File I/O

Introduction

- File I/O refers to **File Input and Output** which is reading data from disk and writing data to files on disk/storage.
- This is a very important aspect of programming for tasks like:
 - Processing user data stored in hard disk.
 - Storing Application Data.
 - Loading a previous state of the application by reading from storage.
 - Reading Configuration Files.

Reading/Writing files in java

There are few different ways to read/write files in Java. We'll be focusing on only one approach in this slide.

To read/write from a file we need to:

- Create a **FileReader/FileWriter object** to read/write the file.
- Then wrap it with a **BufferedReader/BufferedWriter object** for efficient reading/writing.

Examples are given in the next slides

Reading files in java

```
1. import java.io.FileReader;
2. import java.io.BufferedReader;
3. import java.io.IOException;

4. class ReadingFile{
5.     public static void main(String[] args) throws IOException{
6.         String location = "C:\\users\\bracu\\Hello.txt";
7.         FileReader fr = new FileReader(location);
8.         BufferedReader br = new BufferedReader(fr);
9.         String first_line = br.readLine();
10.        String second_line = br.readLine();
11.        String third_line = br.readLine();
12.        System.out.println(first_line);
13.        System.out.println(second_line);
14.        System.out.println(third_line);
15.        br.close();
16.        fr.close();
17.    }
18. }
```

A file called [Hello.txt](#) contains 3 sentences in 3 separate lines.

Output:

Welcome to DrJava.

>

This is the first sentence.
This is the second sentence.
This is the third sentence.

Using loops to read lines from a file

```
1. import java.io.FileReader;
2. import java.io.BufferedReader;
3. import java.io.IOException;
4. class ReadingFile_Loop{
5.     public static void main(String[] args) throws IOException{
6.         String location = "C:\\users\\bracu\\names.txt";
7.         FileReader fr = new FileReader(location);
8.         BufferedReader br = new BufferedReader(fr);
9.         String line = br.readLine();
10.        int i = 1;
11.        while(line!=null){
12.            System.out.println("Name#" + i + ": " + line);
13.            i++;
14.            line = br.readLine();
15.        }
16.        br.close();
17.        fr.close();
18.    }
19. }
```

A file called [names.txt](#) contains unknown amount of names in separate lines.

Output:

```
Welcome to DrJava.
>
Name#1: Hayayo Miyazaki
Name#2: Daenerys Targaryen
Name#3: Chester Bennington
Name#4: Hideo Kojima
Name#5: Amy Lee
Name#6: Rustin Cohle
Name#7: Walter White
Name#8: Luke Skywalker
Name#9: Alan Moore
Name#10: Terry Davis
```

Writing files in java

```
1. import java.io.FileWriter;
2. import java.io.BufferedWriter;
3. import java.io.IOException;
4. class WritingFile{
5.     public static void main(String[] args) throws IOException{
6.         String location = "C:\\\\users\\\\bracu\\\\test.txt";
7.         FileWriter fw = new FileWriter(location);
8.         BufferedWriter bw = new BufferedWriter(fw);
9.
10.        bw.write("First Line\n");
11.        bw.write("Second Line\n");
12.        bw.write("Third Line\n");
13.        bw.close();
14.        fw.close();
15.    }
16. }
17. /*after execution a test.txt file
18. will be created with these lines written.*//
```

Creating a file called **test.txt** and adding 3 separate lines to it.

Using loops to write lines to a file

```
1. import java.io.FileWriter;
2. import java.io.BufferedWriter;
3. import java.io.IOException;

4. class WritingFile_Loop{
5.     public static void main(String[] args) throws IOException{
6.         String[] words = {"Hi", "Hey", "Hello", "Yes", "No"};
7.         String location = "C:\\\\users\\\\bracu\\\\words.txt";
8.         FileWriter fw = new FileWriter(location);
9.         BufferedWriter bw = new BufferedWriter(fw);
10.        for(int i=0; i<words.length; i++) {
11.            bw.write( "Name"+(i+1)+words[i]+"\\n" );
12.        }
13.        bw.close();
14.        fw.close();
15.    }
16. }

17. /*after execution a words.txt file
18. will be created with the words written.*/
```

An array of Strings called **words** is given. Creating a file called words.txt and storing the values from the array in the file.

THANK YOU!